

Roberto Vercellino

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Professional Summary

Energy systems research engineer and quantitative analyst with 5 years of experience developing decision-support models, optimization software, and techno-economic analyses to guide infrastructure and investment decisions for government agencies, utilities, and industry partners. Specialized in energy and power systems modeling, mathematical optimization, HPC-scale simulation, and financial/techno-economic analysis (NPV, IRR, LCOE) to evaluate complex investments under uncertainty. Experienced leading multidisciplinary research and consulting projects end-to-end – from scoping and proposal development to execution and delivery – and translating rigorous quantitative analysis into strategic recommendations for executives, policymakers, and technical stakeholders. Originally from Italy, I earned a basketball scholarship to study in the United States, where I began my professional career before relocating back to Milan. Fluent English and Italian speaker.

Profile & Skills

- **Energy Systems & Markets:** Energy infrastructure planning, decarbonization, renewables, biofuels, nuclear energy, hydrogen, CCS&U, transportation electrification, EV charging, microgrids, distributed energy resources (DERs), battery energy storage systems (BESS), AI data centers, power systems and grid distribution, electricity markets (intraday, balancing, ancillary services), algorithmic bidding and automated real-time dispatch, energy arbitrage.
- **Techno-Economic & Financial Analysis:** Discounted cash flow (DCF), investment appraisal (NPV, IRR, ROI, LCOE), asset valuation, business case development, scenario and sensitivity analysis, risk assessment and mitigation, business planning.
- **Quantitative Decision Analysis & Optimization:** Decision-support modeling, mathematical optimization (linear, nonlinear, mixed-integer, stochastic, simulation-based, derivative-free, surrogate-based programming), model predictive control, Monte Carlo simulation, uncertainty quantification, forecasting, statistical analysis.
- **Programming & Software Engineering:** Python (Pyomo, Pandas, NumPy, SciPy, SimPy, Scikit-Learn, PyTorch), MATLAB/Simulink, SQL, C/C++, Java, Bash, LabVIEW; solvers (Gurobi, CPLEX, Ipopt, CBC); version control (Git), code review, testing (pytest), CI/CD (GitHub Actions), Docker, technical documentation (Sphinx, MkDocs).
- **HPC & Scalable Computing:** SLURM, MPI, multiprocessing, Apptainer containerization, large-scale parallel data processing, simulation and scenario analysis on high-performance computing systems.
- **Machine Learning & AI:** Gradient-Boosted Trees, Random Forest, Gaussian Process, K-Means Clustering, feature engineering, AI workload benchmarking (MLPerf, vLLM), familiarity with distributed computing frameworks (e.g., Spark).
- **Project Leadership & Delivery:** Project scoping, proposal development, budget management, multidisciplinary team leadership and mentorship, client relationship management, project delivery.
- **Stakeholder Engagement & Communication:** C-suite and executive presentations, technical reporting, client workshops, stakeholder interviews, strategic briefings, congressional and regulatory reporting, big-picture narrative development.

Education

Master of Science | *Mechanical Engineering* 2020 – 2022
Colorado State University: 4.0 GPA Fort Collins, Colorado, U.S.A.

Bachelor of Science | *Physics* 2016 – 2020
Regis University: 4.0 GPA Denver, Colorado, U.S.A.

Work Experience

Researcher III, Quantitative Decision Analysis & Optimization 2024 – 2026
Center for Integrated Mobility Sciences, National Laboratory of the Rockies (NLR) [1] Golden, Colorado, U.S.A.

– **Athena Mobility**, U.S. Department of Energy, Vehicle Technology Office 2023 – 2026

- Developed transportation demand scenarios and infrastructure models to advise major U.S. airports on decarbonization strategies, informing **$100+ million in long-term infrastructure investment decisions** and culminating in a briefing to the U.S. Secretary of Energy [1].
- Served as primary analytical point-of-contact to executives from **10 of the largest U.S. airports**, conducting stakeholder interviews, assessing operational constraints, and identifying cost-effective infrastructure investment opportunities through the Aeroportal.
- Developed EVI-Rental, an event-driven Python simulation model estimating costs, infrastructure requirements, and load-flexibility opportunities from EV charging at rental car centers.
- Built a parallel HPC workflow to run more than **5,000 Monte Carlo simulations**, quantifying tradeoffs between up-front infrastructure investments and decarbonization targets and identifying pathways achieving up to **41% in capital savings** [2].

– **EDGES Microgrid Optimization Model**, U.S. Department of Energy 2022 – 2026

- Led technical development of the EDGES decision-support platform for long-term planning of distributed energy resource investments under evolving market, regulatory, and operational conditions [3].
- Developed stochastic optimization algorithms (Pyomo, CBC, Gurobi, Ipopt) for real-time economic dispatch and energy arbitrage of distributed energy resources, minimizing lifecycle costs and grid upgrade requirements [4].
- Refactored the microgrid design-optimization layer, **reducing solution time from 48 hours to 2 hours (24×)** by testing 5 derivative-free and surrogate-based optimization methods [5].
- **Delivered investment-focused recommendations across 10+ projects** for public-sector and industry clients, translating complex modeling outputs into actionable planning insights.

– **Data Center Energy Modeling**, NLR Strategic Funding 2025 – 2026

- **Led a multidisciplinary team of 7 researchers** to develop a first-of-its-kind, bottom-up model of AI data center electricity demand, supporting grid infrastructure planning and capacity expansion studies [6].
- Directed a team of six in adapting AI benchmark workloads (MLPerf, vLLM) and ran HPC experiments measuring server-level power consumption of representative AI workloads (training, fine-tuning, inference).
- Designed and executed reproducible experiments to address critical evidence gaps in AI infrastructure planning, publishing an open dataset and analysis to support industry and policy decision-making [7].

– **Microgrid EMT Simulation Models**, Industry-funded 2025 – 2026

- **Scoped, negotiated, and led delivery** of an industry-funded project developing MATLAB/Simulink electromagnetic transient (EMT) models capturing sub-second interactions between AI data centers, EV fast charging, and microgrid assets.
- **Recruited and coordinated a multidisciplinary team of 10 researchers**, managed project budgets and deliverables, and maintained ongoing engagement with the sponsoring organization throughout execution.

– **Grid Impacts of Electrification at U.S. Army Bases**, U.S. Department of Defense 2025

- Advised the U.S. Army on fleet electrification investment strategies across three military installations, simulating operation, grid impacts, and costs of five electrification scenarios using vehicle telematics and utilization data.
- Integrated stakeholder interviews and sensitivity analysis to identify robust, cost-effective infrastructure pathways. **Co-authored findings within a report submitted to the U.S. Congress**.

[1] Until 2025, NLR was named and known as the *National Renewable Energy Laboratory (NREL)*.

– **HPC Analytics**, NLR Strategic Funding 2025 – 2026

- Supported the ontology design and machine-readable mapping of power flows, telemetry, and control points to enable agentic AI orchestration of NLR's HPC system.

Researcher II, Quantitative Decision Analysis & Optimization 2022 – 2024

Center for Integrated Mobility Sciences, National Renewable Energy Laboratory (NREL) Golden, Colorado, U.S.A.

– **Clean Transportation Program (CTP) Benefits Analysis**, California Energy Commission (CEC) 2023 – 2024

- **Led a 6-person team** evaluating the environmental and socioeconomic impacts of more than **1,000 publicly funded transportation projects** representing **$1.5 billion in investments**, supporting future portfolio and funding strategy for the CEC; results presented to CEC Commissioner Monahan [8, 9].
- Developed investment evaluation frameworks to quantify environmental and socioeconomic KPIs (e.g., CO_2 reduced/$ invested) across alternative funding strategies, reducing uncertainty and supporting risk-informed decision-making.
- Identified, integrated, and validated relevant data from 20+ NREL, CEC, and public sources; automated the ETL cleaning and processing of 20+ spreadsheets of assumption parameters and project data.

– **Public Transit Electrification Study**, New York Metropolitan Transportation Authority (MTA) 2023 – 2024

- Advised the New York MTA on public transit electrification strategies, simulating operation of an electrified bus depot to evaluate infrastructure requirements, charging loads, projected costs, and impact-mitigation opportunities using onsite battery storage and PV [10].

– **Microgrid Market Analysis**, Industry-funded 2022 – 2025

- Conducted large-scale sensitivity analysis to evaluate how market conditions, policy environments, and technology adoption trends influence the economic value and risk profile of distributed energy investments.
- Developed a **web application to process and visualize 58,000 U.S. utility tariffs** (SQL) from NREL's Utility Rate Database, enabling stakeholders to identify high-value markets for EV charging, microgrid, and renewable energy investments.

Graduate Research Assistant 2020 – 2022

Herber Research Group, Colorado State University Fort Collins, Colorado, U.S.A.

– **FLECCS Project**, U.S. Department of Energy, ARPA-E Program

- Constructed MATLAB dynamic models of natural gas combined-cycle power plants integrated with carbon capture and thermal energy storage [11].
- Implemented nested optimization to jointly determine optimal system design and control, maximizing economic value (NPV, LCOE) through energy arbitrage.
- Evaluated the economic performance and CO_2 emissions of more than 20 technology configurations across hundreds of current and future electricity market scenarios.
- Delivered weekly update presentations to the team and contributed to 5 quarterly meetings with ARPA-E.

Laboratory Technician & Data Analysis Assistant 2019

MoVeIT Research Group, Medical Physics Department, University of Torino Torino, Italy

– **MoVeIT Project**, Centro Nazionale di Adroterapia Oncologica (CNAO)

- Designed and executed experiments (LabVIEW) to evaluate the performance and reliability of prototype instrumentation for hadron (cancer) therapy applications, including noise characterization of a custom ASIC interfaced with a prototype proton sensor.
- Performed offline data analysis and curve fitting to quantify measurement uncertainty, assess system performance, and identify key limitations.

Journal Articles & Reports

[5] Campos, G. et al. "Parallel derivative-free optimization for simulation-based design of behind-the-meter energy systems". In: *Computers & Chemical Engineering* 205 (2025), p. 109422. ISSN: 0098-1354. DOI: 10.1016/j.compchemeng.2025.109422.

[6] Vercellino, R. et al. "Measurement of Generative AI Workload Power Profiles for Whole-Facility Data Center Infrastructure Planning". In: *arXiv* arXiv:2604.07345 (Apr. 8, 2026). DOI: 10.48550/arXiv.2604.07345.

[8] Vercellino, R. et al. *Benefits Associated with Projects and Technologies Supported by the Clean Transportation Program: 2023 Analysis*. CEC-600-2025-023. California Energy Commission, June 18, 2025. URL: https://www.energy.ca.gov/publications/2025/benefits-associated-projects-and-technologies-supported-clean-transportation.

[12] He, Y. et al. "Energy, power, and infrastructure demands from electrifying airport ground support equipment at United States airports". In: *Nature Communications* (Mar. 30, 2026). ISSN: 2041-1723. DOI: 10.1038/s41467-026-71125-4.

[14] Limb, B. J. et al. "Economic viability of using thermal energy storage for flexible carbon capture on natural gas power plants". In: *Journal of Energy Storage* 55 (2022), p. 105836. ISSN: 2352-152X. DOI: 10.1016/j.est.2022.105836.

[15] Markey, E. J. et al. "Economic impact of thermal energy storage on natural gas power with carbon capture in future electricity markets". In: *International Journal of Greenhouse Gas Control* 133 (2024), p. 104098. ISSN: 1750-5836. DOI: 10.1016/j.ijggc.2024.104098.

[21] Huyett, J. et al. *Synergistic Heat Pumped Thermal Storage and Flexibly Carbon Capture System*. Colorado State Univ., Fort Collins, CO (United States), 2023. URL: https://doi.org/10.2172/2202638.

Conference Proceedings

[2] Vercellino, R. et al. "Behind-the-meter Energy Storage and Generation in Support of Electrified Rental Car Centers". In: *2025 IEEE Electrical Energy Storage Applications and Technologies Conference (EESAT)*. 2025, pp. 1–5. DOI: 10.1109/EESAT62935.2025.10891234.

[3] Campos, G. et al. "Optimal Design and Control of Behind-the-Meter Resources for Retail Buildings with EV Fast Charging". In: *Systems and Control Transactions*. Vol. 3. 2024, pp. 417–425. DOI: 10.69997/sct.150240.

[4] Campos, G. et al. "Impact of load uncertainty from electrified rental car facilities on the control and design of behind-the-meter battery storage and PV generation". In: *2025 American Control Conference (ACC)*. 2025, pp. 766–771. DOI: 10.23919/ACC63710.2025.11107983.

[11] Vercellino, R. et al. "Control Co-Design Optimization of Natural Gas Power Plants With Carbon Capture and Thermal Storage". In: *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC-CIE)*. Vol. 3A: 48th Design Automation Conference (DAC). Aug. 2022. DOI: 10.1115/DETC2022-90021.

[16] Limb, B. et al. "Evaluating the economic viability of NGCC-SWITCC: Natural Gas Combined Cycle System With Integrated Thermal storage and Carbon Capture". In: *Proceedings of the 16th Greenhouse Gas Control Technologies Conference (GHGT-16)*. 2022, pp. 23–24. DOI: 10.2139/ssrn.4295563.

Patents

[22] Bandhauer, T. M. et al. "Synergistic heat pumped thermal storage and flexible carbon capture system". Pat. U.S. Patent US20230392523A1. 2021.

Master's Thesis

[19] Vercellino, R. "Analysis and Control Co-design Optimization of Natural Gas Power Plants with Carbon Capture and Thermal Energy Storage". Colorado State University, 2022. URL: https://www.proquest.com/openview/4f6664f58e089ac4947ec795c87ca1de/1?pq-origsite=gscholar&cbl=18750&diss=y.

Datasets, Presentations & Other Research Outputs

[1] Vercellino, R. et al. *Athena ZEV Overview*. Presented at: EPRI Infrastructure Working Council. June 2024. URL: https://publicdownload.epri.com/PublicAttachmentDownload.svc/AttachmentId=88883.

[7] Vercellino, R. et al. *Dataset of Generative AI Workload Power Profiles*. Jan. 30, 2026. DOI: 10.7799/3025227.

[9] Vercellino, R. et al. *Analysis of Benefits Associated With Projects and Technologies Supported by the Clean Transportation Program: 2023 Executive Summary [Slides]*. 2023. URL: https://doi.org/10.2172/2203224.

[10] Vercellino, R. et al. *Evaluation of Behind-the-Meter Resources for Public Transit Electrification: Insights from the NY-MTA*. Presented at: EPRI Bus & Truck Working Council. Oct. 2023. URL: https://publicdownload.epri.com/PublicAttachmentDownload.svc/AttachmentId=86072.

[13] He, Y. et al. *Airport Ground Support Equipment Analysis*. Apr. 7, 2026. DOI: 10.15483/3014721.

[17] Vercellino, R. et al. *Control Co-Design Optimization of Natural Gas Power Plants With Carbon Capture and Thermal Storage*. Presented at: International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. Aug. 2022. URL: https://www.engr.colostate.edu/%7Edrherber/files/Vercellino2022c_presentation.pdf.

[18] Vercellino, R. et al. *Realizing Profitable Carbon Capture on Natural Gas Plants with Thermal Energy Storage through Optimization-Based Decision Support*. Presented at: International Symposium on Sustainable Systems and Technology. June 2022. URL: https://www.engr.colostate.edu/~drherber/files/Vercellino2022a.pdf.

[20] Vercellino, R. *Analysis and Control Co-design Optimization of Natural Gas Power Plants with Carbon Capture and Thermal Energy Storage*. Master's Thesis Defense Presentation. 2022. URL: https://www.engr.colostate.edu/%7Edrherber/files/Vercellino2022b_presentation.pdf.

TEACHING

Undergraduate Teaching Assistant 2019
Department of Physics & Astronomy, Regis University Denver, Colorado, U.S.A.
- Assisted with intro-level Physics with Calculus laboratory.

HONORS & AWARDS

NREL High-Impact Business Development Award 2025
Internal recognition for securing $800,000 in industry-funded research projects.

NREL Key Contributor Award 2023, 2025
Internal award recognizing exceptional research contributions and project impact.

Best Poster at ISSST Academic Conference 2022
Awarded best poster for research on optimization-based carbon capture and thermal energy storage systems.

Reviewer for the ASME IDETC-CIE, ISSST, and IEEE-EESAT Academic Conferences 2022 – 2025
Reviewed conference submissions for research quality and methodological soundness.

Physics & Astronomy Graduating Student of the Year 2020
Recognition of academic success from Regis University.

CoSIDA Academic All-District First Team 2020
NCAA recognition for academic and basketball success.

Full College Athletic Scholarship Awarded for 4 Years 2016 – 2020
Managed consistent academic success while competing in the NCAA basketball league.

VOLUNTEERING & COMMUNITY INVOLVEMENT

High School Student Mentor 2023 – 2026
Denver Public Schools Denver, Colorado, U.S.A.
- Supported juniors and seniors through career exploration in the sustainable energy field.

Physics Station Coordinator 2019
Regis University Science Sunday Denver, Colorado, U.S.A.
- Coordinated laboratories to promote physics and its applications to more than 150 children.

Basketball Station Coordinator 2019
Special Olympics Games Westminster, Colorado, U.S.A.
- Organized and supervised sport games to foster sports to children as carrier of values like teamwork and social inclusivity.

LANGUAGES

Italian Native

English Fluent

Personal Interests & Hobbies

Hiking & Camping
Hiked and summited 10 of Colorado's 58 peaks above 14,000 feet (4,267.2 meters), also known as 14ers.

Fermented Foods & Beverages
Regularly bake sourdough bread and pizza, and brew homemade kombucha tea.

Basketball
Played basketball in Italy's top professional league and at the NCAA D-I level in college.